

# SYLLABUS

## (ACADEMIC SESSION 2014-15)

### FOUNDATION OF COMPUTER SCIENCE

#### INSTRUCTIONS TO PAPER SETTERS

Maximum Marks: 75

1. Question No. 1 should be compulsory and cover the entire syllabus. This question should have objective or short answer type questions. It should be of 25 marks.

2. Apart from Question No. 1, the rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions. However, the student may be asked to attempt only 1 question from each unit. Each question should be 12.5 marks.

#### UNIT-I

**Formal Logic:** Proposition, Symbolic Representation and logical entailment theory of Inferences and tautologies, Predicates, Quantifiers, Theory of inferences for predicate calculus, resolution. Techniques for theorem proving: Direct Proof, Proof by Contraposition, proof by contradiction. [T1, T2] [No. of hrs. 10]

#### UNIT-II

**Overview of Sets and set operations, permutation and combination, principle of inclusion, exclusion (with proof) and pigeonhole principle (with proof), Relation, operation and representation of a relation, equivalence relation, POSET, Hasse Diagrams, extremal Elements, Lattices, composition of function, inverse, binary and n-ary operations.** [T1, T2] [No. of hrs. 12]

#### UNIT-III

**Principle of mathematical induction, principle of complete induction, solution methods for linear and non-linear first-order recurrence relations with constant coefficients, Graph Theory: Terminology, isomorphic graphs, Euler's formula (proof), chromatic number of a graph, five color theorem(with proof), Euler & Hamiltonian paths.** [T1, T2] [No. of hrs. 11]

#### UNIT-IV

**Groups, Symmetry, subgroups, normal subgroups, cyclic group, permutation group and Cayley's theorem(without proof), cosets Lagrange's theorem(with proof) homomorphism, isomorphism, automorphism, rings, Boolean function, Boolean expression, representation & minimization of Boolean function.** [T1, T2] [No. of hrs. 11]

## NEW TOPICS ADDED FROM ACADEMIC SESSION [2022-23] THIRD SEMESTER [B.TECH] DISCRETE MATHEMATICS (CIC-205)

⇒ Functions, Growth of functions, Permutation functions, Partially ordered sets

#### [UNIT-2]

**Q.1 Prove a function  $f: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = 2x - 3$  is a bijective function.**

**Ans.** We have to prove this function is both injective and surjective.

If  $f(x_1) = f(x_2)$

then  $2x_1 - 3 = 2x_2 - 3$  and it implies that  $x_1 = x_2$ .

Hence,  $f$  is injective.

Here,  $2x - 3 = y$

So,  $x = (y + 3)/2$  which belongs to  $\mathbb{R}$  and  $f(x) = y$ .

Hence,  $f$  is surjective.

Since  $f$  is both surjective and injective, we can say  $f$  is bijective.

**Q.2 Let  $f(x) = x + 2$  and  $g(x) = 2x + 1$ , find  $(f \circ g)(x)$  and  $(g \circ f)(x)$ .**

**Ans.**  $(f \circ g)(x) = f(g(x)) = f(2x + 1) = 2x + 1 + 2 = 2x + 3$

$(g \circ f)(x) = g(f(x)) = g(x + 2) = 2(x + 2) + 1 = 2x + 5$

Hence,  $(f \circ g)(x) \neq (g \circ f)(x)$ .

**Q.3 Show that the function  $f: \mathbb{R} \rightarrow \mathbb{R}$ , given by  $f(x) = 2x$ , is one-one and onto.**

**Ans.** For one-one:

Let  $a, b \in \mathbb{R}$  such that  $f(a) = f(b)$  then,

$f(a) = f(b)$

$\Rightarrow 2a = 2b$

$\Rightarrow a = b$

Therefore,  $f: \mathbb{R} \rightarrow \mathbb{R}$  is one-one.

For onto:

Let  $p$  be any real number in  $\mathbb{R}$  (co-domain).

$f(x) = p$

$\Rightarrow 2x = p$

$\Rightarrow x = p/2$

$p/2 \in \mathbb{R}$  for  $p \in \mathbb{R}$  such that  $f(p/2) = 2(p/2) = p$

For each  $p \in \mathbb{R}$  (codomain) there exists  $x = p/2 \in \mathbb{R}$  (domain) such that  $f(x) = y$

For each element in codomain has its pre-image in domain.

So,  $f: \mathbb{R} \rightarrow \mathbb{R}$  is onto.

Since  $f: \mathbb{R} \rightarrow \mathbb{R}$  is both one-one and onto.

$f: \mathbb{R} \rightarrow \mathbb{R}$  is one-one correspondent (bijective function).

**Q.4 If  $f: \mathbb{R} \rightarrow \mathbb{R}$ :  $f(x) = 3x + 5$  is a bijective function then, find the inverse of  $f$ .**

**Ans.** Let  $x \in \mathbb{R}$  (domain),  $y \in \mathbb{R}$  (codomain) such that  $f(x) = y$

$f(x) = y$

$$\Rightarrow 3x + 5 = y$$

$$\Rightarrow x = (y - 5)/3$$

$$\Rightarrow f^{-1}(y) = (y - 5)/3$$

Thus,  $f^{-1}: R \rightarrow R$  is defined as  $f^{-1}(x) = (x - 5)/3$  for all  $x \in R$ .

**Q.5** Let  $f: R \rightarrow R$ ;  $f(x) = \cos x$  and  $g: R \rightarrow R$ ;  $g(x) = x^3$ . Find  $f \circ g$  and  $g \circ f$ .

**Ans.** Since the range of  $f$  is a subset of the domain of  $g$  and the range of  $g$  is a subset of the domain of  $f$ . So,  $f \circ g$  and  $g \circ f$  both exist.

$$g \circ f(x) = g(f(x)) = g(\cos x) = (\cos x)^3 = \cos^3 x$$

$$f \circ g(x) = f(g(x)) = f(x^3) = \cos x^3$$

**Q.6** If  $f: Q \rightarrow Q$  is given by  $f(x) = x^2$ , then find  $f^{-1}(25)$ .

**Ans.** Let  $f^{-1}(25) = x$

$$f(x) = 25$$

$$x^2 = 25$$

$$x = \pm 5$$

$$f^{-1}(25) = \{-5, 5\}$$

**Q.7** Consider the function  $f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$  given by

$$f(n) = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 4 \end{pmatrix}$$

(a) Find  $f(1)$ .

(b) Find an element  $n$  in the domain such that  $f(n) = 1$ .

(c) Find an element  $n$  of the domain such that  $f(n) = n$ .

(d) Find an element of the codomain that is not in the range.

**Ans.**  $f(1) = 4$ , Since 4 is the number below 1 in the two-line notation.

Such an  $n$  is,  $n = 2$ , since  $f(2) = 1$ . Note that 2 is above a 1 in the notation.

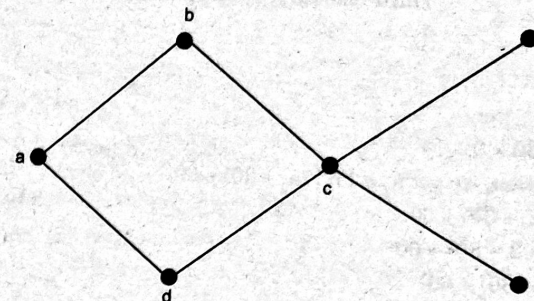
$n = 4$  has this property. We say that 4 is a fixed point of  $f$ . Not all function have such a point.

Such an element is 2 (in fact, that is the only element in the codomain that is not in the range). In other words, 2 is not the image of any element under  $f$ ; nothing is sent to 2.

**Q.8** The coach of a basketball team is picking among 11 players for the 5 different positions in his starting lineups. How many different lineups can he pick?

**Ans.**  $P(11, 5) = \frac{11!}{(11-5)!} = \frac{11!}{6!} = 55440$  different lineups. (The permutation function is defined as:  $P(n, k) = \frac{n!}{(n-k)!}$ )

**Q.9** Determine all the maximal and minimal elements of the poset whose Hasse diagram is shown in fig.



**Ans.** The minimal elements of the Poset are  $d$  and  $c$ .

The maximal elements of Poset are  $b$  and  $f$ .

**Q.10** Consider  $A = \{1, 2, 3, 5, 6, 10, 15, 30\}$  is ordered by divisibility. Determine all the comparable and non-comparable pairs of elements of  $A$ .

**Ans.** The comparable pairs of elements of  $A$  are:

$\{1, 2\}, \{1, 3\}, \{1, 5\}, \{1, 6\}, \{1, 10\}, \{1, 15\}, \{1, 30\}$

$\{2, 6\}, \{2, 10\}, \{2, 30\}$

$\{3, 6\}, \{3, 15\}, \{3, 30\}$

$\{5, 10\}, \{5, 15\}, \{5, 30\}$

$\{6, 30\}, \{10, 30\}, \{15, 30\}$

The non-comparable pair of elements of  $A$  are:

$\{2, 3\}, \{2, 5\}, \{2, 15\}$

$\{3, 5\}, \{3, 10\}, \{5, 6\}, \{6, 10\}, \{6, 15\}, \{10, 15\}$

Note: In an ordered set  $A$ . Two elements  $a$  and  $b$  of set  $A$  are called comparable if  $a \leq b$  or  $b \leq a$  in a relation. Whereas two elements  $a$  and  $b$  of set  $A$  are called noncomparable if neither  $a \leq b$  nor  $b \leq a$ .

### RECURRENCE RELATION [UNIT-2]

**Q.11** Consider the recurrence relation  $a_1 = 4, a_n = 5n + a_{n-1}$ . What is the value of  $a_{64}$ ?

$$\begin{aligned} \text{Ans. } a_n &= 5n + a_{n-1} \\ &= 5n + 5(n-1) + \dots + a_{n-2} \\ &= 5n + 5(n-1) + 5(n-2) + \dots + a_1 \\ &= 5n + 5(n-1) + 5(n-2) + \dots + 4 \text{ [since, } a_1 = 4\text{]} \\ &= 5n + 5(n-1) + 5(n-2) + \dots + 5.1 - 1 \\ &= 5(n + (n-1) + \dots + 2 + 1) - 1 \\ &= 5 \times n(n+1)/2 - 1 \\ a_n &= 5 \times n(n+1)/2 - 1 \end{aligned}$$

Now,  $n = 64$  so the answer is  $a_{64} = 10399$ .

**Q.12** Determine the value of  $a_3$  for the recurrence relation  $a_n - 17a_{n-1} + 30n$  with  $a_0 = 3$ .

**Ans.** For  $n = 1$ ,

$$a_1 = 17a_0 + 30(1)$$

$$a_1 = 17(3) + 30 \text{ since } a_0 = 3$$



$$a_1 = 51 + 30$$

$$a_1 = 81$$

For  $n = 2$ ,

$$a_2 = 17a_1 + 30 \times 2$$

...[2]

By substitution, we get  $a_2 = 17(17a_0 + 30) + 60$ .

[from 2]

$$a_2 = 17(17a_0 + 30) + 60$$

$$a_2 = 17(17 \times 3 + 30) + 60$$

$$a_2 = 17(51 + 30) + 60$$

$$a_2 = 17(81) + 60$$

$$a_2 = 1437$$

...[3]

For  $n = 3$

$$a_3 = 17a_2 + 30 \times 3$$

$$a_3 = 17(1437) + 30 \times 3$$

[from 3]

$$a_3 = 24519$$

Then regrouping the terms, we get  $a_3 = 24519$ , where  $a_0 = 3$ ,  $a_1 = 81$ ,  $a_2 = 1437$ .

⇒ GCD, LCM, Prime numbers [UNIT - 2]

**Q.13 Evaluate the expression:  $(X + Z)(X + XZ') + XY + Y$ .**

**Ans.**  $(X + Z)(X + XZ') + XY + Y$  [Given]

$= (X + Z)X(1 + Z') + XY + Y$  [Using Distributive]

$= (X + Z)X + XY + Y$  [Using Complement, Identity]

$= (X + Z)X + Y(X + 1) \Rightarrow (X + Z)X + Y$  [Using Idempotent]

$= XX + ZX + Y$  [Using Distributive]

$= X + XZ + Y$  [Using Identity]

$= X(1 + Z) + Y$

$= X + Y$  [Idempotent]

**Q.14 What are the canonical forms of Boolean Expressions?**

**Ans.** There are two kinds of canonical forms for a Boolean expression:

- sum of minterms (SOM) form
- Product of maxterms (SOM) form.

**Q.15 Find the prime factorization of each of these integers. (a) 88 (b) 126**

(c) 729 (Unit 2: GCD, LCM, Prime No.)

**Ans. (a)**  $\sqrt{88} \approx 9.38$

$$88/2 = 44$$

$$44/2 = 22$$

$$22/2 = 11 \text{ Therefore } 88 = 2^3 \cdot 11$$

(b)  $\sqrt{126} \approx 11.22$

$$126/2 = 63$$

$$63/3 = 21$$

$$21/3 = 7 \text{ Therefore } 63 = 2 \cdot 3^2 \cdot 7$$

(c)  $\sqrt{729} = 27$

$$729/3 = 243$$

$$243/3 = 81$$

$$81/3 = 27$$

$$27/3 = 9$$

$$9/3 = 9 \text{ Therefore } 729 = 3^6$$

**Q.16 What are the greatest common divisors of these pairs of integers?**

(a)  $3^7 \cdot 5^3 \cdot 7^3, 2^{11} \cdot 3^5 \cdot 5^9$

**Ans.**  $2^{\min(0,11)} \cdot 3^{\min(7,5)} \cdot 5^{\min(3,9)} \cdot 7^{\min(3,0)} = 3^5 \cdot 5^3$

(b)  $11 \cdot 13 \cdot 17 \cdot 2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3$

**Ans.**  $2^{\min(9,0)} \cdot 3^{\min(7,0)} \cdot 5^{\min(5,0)} \cdot 7^{\min(3,0)} \cdot 11^{\min(1,0)} \cdot 13^{\min(1,0)} \cdot 17^{\min(1,0)}$

(c)  $41 \cdot 43 \cdot 53, 41 \cdot 43 \cdot 53$

**Ans.**  $41 \cdot 43 \cdot 53$

(d)  $23^{31} \cdot 23^{17}$

**Ans.**  $23^{\min(31,17)} = 23^{17}$

**Q.17 What is the least common multiple of these pairs of integers?**

(a)  $3^7 \cdot 5^3 \cdot 7^3, 2^{11} \cdot 3^5 \cdot 5^9$

**Ans.**  $2^{\max(0,11)} \cdot 3^{\max(7,5)} \cdot 5^{\max(3,9)} \cdot 7^{\max(3,0)} = 2^{11} \cdot 3^7 \cdot 5^9 \cdot 7^3$

(b)  $11 \cdot 13 \cdot 17 \cdot 2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3$

**Ans.**  $11 \cdot 13 \cdot 17 \cdot 2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3$

(c)  $23^{31}, 23^{17}$

**Ans.**  $23^{\max(31,17)} = 23^{31}$

(d)  $41 \cdot 43 \cdot 53, 41 \cdot 43 \cdot 53$

**Ans.**  $41 \cdot 43 \cdot 53$

**Q.18 Find gcd(92928, 123552) and lcm(92928, 123552) and verify that gcd(92928, 123552) · lcm(92928, 123552) = 92928 · 123552. [Hint: First find the prime factorizations of 92928 and 123552].**

**Ans.**  $92928 = 2^8 \cdot 3 \cdot 11^2$

$$123552 = 2^5 \cdot 3^3 \cdot 11 \cdot 13$$

$$\text{gcd}(92928, 123552) = 2^5 \cdot 3 \cdot 11$$

$$\text{lcm}(92928, 123552) = 2^8 \cdot 3^3 \cdot 11^2 \cdot 13$$

$$\text{gcd}(92928, 123552) \cdot \text{lcm}(92928, 123552) = 92928 \cdot 123552$$

$$(2^5 \cdot 3 \cdot 11) \cdot (2^8 \cdot 3^3 \cdot 11^2 \cdot 13) = (2^8 \cdot 3 \cdot 11^2) \cdot (2^5 \cdot 3^3 \cdot 11 \cdot 13)$$

$$2^{13} \cdot 3^4 \cdot 11^3 \cdot 13 = 2^{13} \cdot 3^4 \cdot 11^3 \cdot 13$$

**Solution method for divide and conquer recurrence relation, Masters theorem (with proof)**

[UNIT - 2]

**Q.19 Suppose that  $f(n) = f(n/5) + 3n^2$  when  $n$  is a positive integer divisible by 5, and  $f(1) = 3$ . Find (a)  $f(5)$ , (b)  $f(3125)$ .**

**Ans. (a)**  $f(5) = f(5/5) + 3(5)^2$

$$= f(1) + 3(25) \quad \text{since } f(1) = 3$$

$$= 3 + 75 = 78$$

(b)  $f(3125)$ 

$$f(625) = f(625/5) + 3(625)^2$$

$$= f(125) + 3(390625)$$

$$= 48829 + 1171875$$

$$= 1220704$$

$$f(3125) = f(3125/5) + 3(3125)^2$$

$$= f(625) + 3(9765625)$$

$$= 1220704 + 29296875 = 30517579$$

(Note: Master Theorem: Let  $f$  be an increasing function that satisfies the recurrence relation

$f(n) = af(n/b) + cn^d$  whenever  $n = b^k$ , where  $k$  is a positive integer,  $a \geq 1$ ,  $b$  is an integer greater than 1 and  $c$  and  $d$  are real numbers with  $c$  positive and  $d$  non-negative. Then

$$f(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d \log n) & \text{if } a = b^d \\ O(n^{\log_b a}) & \text{if } a > b^d \end{cases}$$

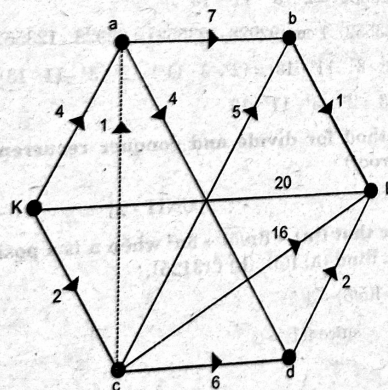
**Q.20** Given a big-O estimable for the function  $f(n) = 2f(n)/3 + 4$  if  $f$  is an increasing function and  $n = 3^k$ .

**Ans.** Use Master Theorem with  $a = 2$ ,  $b = 3$ ,  $c = 4$ ,  $d = 0$ . Since  $a > b^d$ , we know that  $f(n)$  is  $O(n^{\log_3 2}) = O(n^{\log_3 2})$ .

$\Rightarrow$  **Shortest path and Minimum spanning trees and algorithms of the graph MST.**

## [UNIT-4]

**Q.21** Find the shortest paths between K and L in the graph shown in fig using Dijkstra's Algorithm.



**Ans.** To find the Shortest paths between the graph follow steps:

(a) Remove all the loops.

(b) Remove the parallel edges keeping only the edge carrying least weight.

(i) **Determine all the direct paths from K to all other vertices without going through any other vertex.**

| Vertices | K | a | b        | c | D        | L        |
|----------|---|---|----------|---|----------|----------|
| K        | 0 | 4 | $\infty$ | 2 | $\infty$ | 20       |
| a        | 0 | 0 | 7        | 1 | 4        | $\infty$ |
| b        | 0 | 0 | 0        | 0 | 0        | 1        |
| c        | 0 | 1 | 5        | 0 | 6        | 16       |
| d        | 0 | 0 | 0        | 0 | 0        | 2        |
| L        | 0 | 0 | 0        | 0 | 0        | 0        |

(ii) **Determine shortest paths to all vertices through this vertex and update the values. The closest vertex is c.**

| Vertices | K | a      | b      | c  | D      | L       |
|----------|---|--------|--------|----|--------|---------|
| K        | 0 | 3(K,c) | 7(K,c) | 2K | 8(K,c) | 18(K,c) |

(iii) **The vertex which is 2nd nearest to K is b which is used to determine the shortest paths and update the values.**

| Vertices | K | a      | b      | c  | D        | L       |
|----------|---|--------|--------|----|----------|---------|
| K        | 0 | 3(K,c) | 7(K,c) | 2K | 7(K,c,a) | 18(K,c) |

(iv) **The vertex which is 3rd nearest to K is b, which is used to determine the shortest paths and update the values.**

| Vertices | K | a      | b      | c  | D        | L         |
|----------|---|--------|--------|----|----------|-----------|
| K        | 0 | 3(K,c) | 7(K,c) | 2K | 7(K,c,a) | 18(K,c,b) |

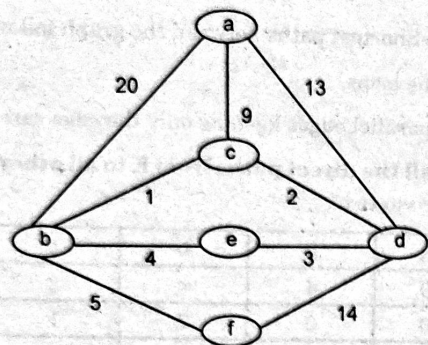
(v) **The vertex which is 3rd nearest to K is d, which is used to determine the shortest paths and update the values.**

| Vertices   | K | A      | b      | c  | D        | L         |
|------------|---|--------|--------|----|----------|-----------|
| K(c,a,b,d) | 0 | 3(K,c) | 7(K,c) | 2K | 7(K,c,a) | 18(K,c,b) |

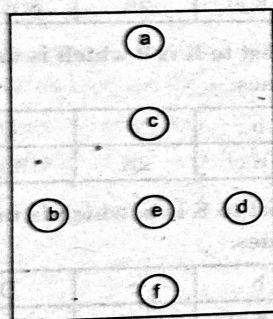
Thus, the shortest distance between K and L is 8 and the shortest path is  $K, c, b, L$ .

**Q.22** Use Prim's algorithms to find minimum spanning tree for the following graph G

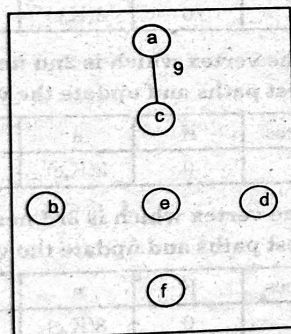


**Ans. Prim's Algorithm**

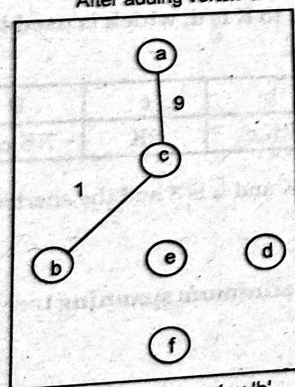
- Initialize the minimal spanning tree with a single vertex, randomly chosen from the graph.
- Repeat steps 3 and 4 until all the vertices are included in the tree.
- Select an edge that connects the tree with a vertex not yet in the tree, so that the weight of the edge is minimal and inclusion of the edge does not form a cycle.
- Add the selected edge and the vertex that connects to the tree.



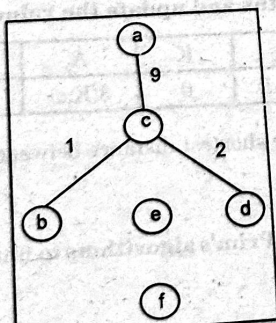
After adding vertex 'a'



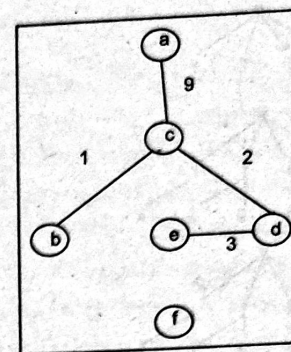
After adding vertex 'c'



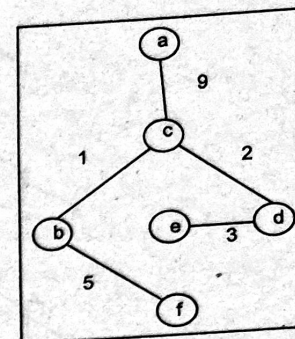
After adding vertex 'b'



After adding vertex 'd'



After adding vertex 'e'



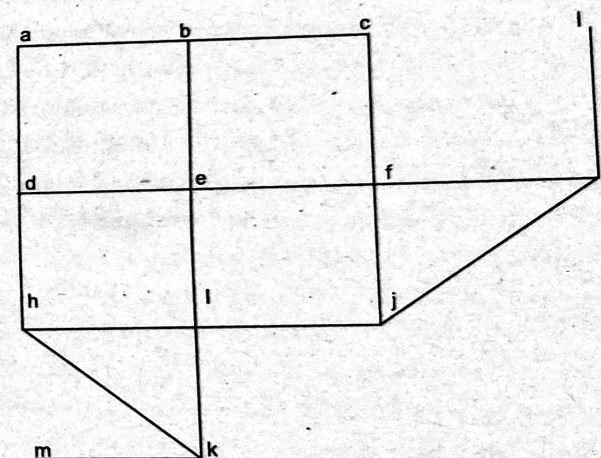
After adding vertex 'f'

The minimum spanning tree is formed carrying the weight as:

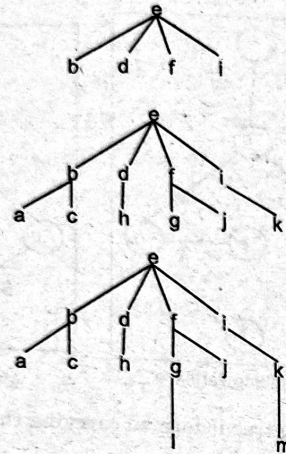
$$1 + 2 + 3 + 5 + 9 = 20$$

**Breadth First search and Depth First Search**

Q.23 Use BFS to find a spanning tree for graph given below.



Ans. Choose root vertex = e



## UNIT - I

- Q.1. Refer to Q.1 (a),(c),(d),(e) of First Term Examination 2014 (Pg. no. 1-2014)
- Q.2. Refer to Q.2 (a),(b) of First Term Examination 2014 (Pg. no. 2-2014)
- Q.3. Refer to Q.3 (a),(b) of First Term Examination 2014 (Pg. no. 2-2014)
- Q.4. Refer to Q.4 (a),(b) of First Term Examination 2014 (Pg. no. 3-2014)
- Q.5. Refer to Q.3 (a) of Second Term Examination 2014 (Pg. no. 8-2014)
- Q.6. Refer to Q.1 (a),(b),(c),(d),(e) of End Term Examination 2014 (Pg. no. 11,12-2014)
- Q.7. Refer to Q.2 (a),(b),(c) of End Term Examination 2014 (Pg. no. 14,15-2014)
- Q.8. Refer to Q.3 (a),(b),(c) of End Term Examination 2014 (Pg. no. 16-2014)
- Q.9. Refer to Q.4 (b),(c) of End Term Examination 2014 (Pg. no. 17,18-2014)
- Q.10. Refer to Q.5 (a),(c) of End Term Examination 2014 (Pg. no. 18,19-2014)
- Q.11. Refer to Q.6 (b) of End Term Examination 2014 (Pg. no. 19-2014)
- Q.12. Refer to Q.1 (a),(b),(c),(d),(e) of First Term Examination 2015 (Pg. no. 1-2015)
- Q.13. Refer to Q.2 (a),(b) of First Term Examination 2015 (Pg. no. 2-2015)
- Q.14. Refer to Q.3 (a),(b) of First Term Examination 2015 (Pg. no. 2,3-2015)
- Q.15. Refer to Q.1 (a) of Second Term Examination 2015 (Pg. no. 5-2015)
- Q.16. Refer to Q.1 (a),(b),(c),(d),(e),(f) of End Term Examination 2015 (Pg. no. 12,13-2015)
- Q.17. Refer to Q.2 (a),(b),(c) of End Term Examination 2015 (Pg. no. 14,15,16-2015)
- Q.18. Refer to Q.3 (a),(b),(c) of End Term Examination 2015 (Pg. no. 16,17-2015)
- Q.19. Refer to Q.4 (c) of End Term Examination 2015 (Pg. no. 18-2015)
- Q.20. Refer to Q.5 (a) of End Term Examination 2015 (Pg. no. 18-2015)
- Q.21. Refer to Q.6 (b) of End Term Examination 2015 (Pg. no. 19-2015)
- Q.22. Refer to Q.1 Important Questions (Pg. no. 25-2015)
- Q.23. Refer to Q.2 Important Questions (Pg. no. 25-2015)
- Q.24. Refer to Q.3 Important Questions (Pg. no. 25-2015)
- Q.25. Refer to Q.1 (c) of First Term Examination 2016 (Pg. no. 1-2016)
- Q.26. Refer to Q.2 (a),(b) of First Term Examination 2016 (Pg. no. 2-2016)
- Q.27. Refer to Q.3 (a) of First Term Examination 2016 (Pg. no. 2-2016)
- Q.28. Refer to Q.4 (a),(b) of First Term Examination 2016 (Pg. no. 3,4-2016)
- Q.29. Refer to Q.1 (a),(b),(d) of End Term Examination 2016 (Pg. no. 5,6-2016)
- Q.30. Refer to Q.2 (b),(c) of End Term Examination 2016 (Pg. no. 6,7-2016)
- Q.31. Refer to Q.3 (c) of End Term Examination 2016 (Pg. no. 8-2016)
- Q.32. Refer to Q.6 (c) of End Term Examination 2016 (Pg. no. 11-2016)
- Q.33. Refer to Q.1 (c),(d) of First Term Examination 2017 (Pg. no. 1-2017)
- Q.34. Refer to Q.2 (a),(b) of First Term Examination 2017 (Pg. no. 2-2017)
- Q.35. Refer to Q.3 (a),(b) of First Term Examination 2017 (Pg. no. 2,3-2017)